

Fiduciaries (NCREIF), which collects property data from its members. NCREIF computes a value-weighted index based on appraisals on a quarterly basis. The indirect property returns are the FTSE NAREIT All Equity returns, also at the quarterly frequency. I compare these property returns with S&P 500 equity returns.

Table 11.11 shows that NCREIF returns are very smooth (the quarterly autocorrelation is 78% and volatility is just 2.2%) compared to REIT and equity returns, which have autocorrelations close to zero and volatilities around 8% to 9%. Notably, the correlation between REIT and equity returns is high, at 63%, while the correlation between NCREIF and REIT returns is low, at 15%. Thus REITs resemble equities more than they resemble direct real estate. Hartzell and Mengden, who were industry practitioners at Salomon Brothers, were the first to document this phenomenon in 1986, and they generated substantial discussion about the underlying nature of real estate returns. Was real estate indeed a separate asset class, or was it just a different version of equity (and debt)?

This debate continues in real estate economics today. My takeaways for asset owners from this long literature are:

Direct Real Estate Returns Are Not Returns

The NCREIF series (and the corresponding widely used Investment Property Databank series in Europe) have many data biases because the values are not based on market transactions. REIT returns, in contrast, are investable returns. There are two important and large biases:³³

1. Smoothing bias

Using appraisals (done at most once or twice a year) artificially induces smoothness,³⁴ which is why the autocorrelation of the NCREIF series is so much higher than REITs or equities in Table 11.11.

2. Selection bias

The properties that you see sold are not representative of the entire stock. For example, perhaps only the best properties are sold because those are the ones a developer is sprucing up, while the foreclosed crummy ones are not being sold.³⁵

³³ There are two minor biases in addition to these two main ones. The first is an index construction bias, which is that repeat-sales indexes, which are developed to deal with appraisal bias, are equal-weighted cross-sectional estimators. But returns of equal-weighted portfolios are arithmetic averages of cross-sectional individual asset returns. I warned you that this was technical! The second is that the time a property is on the market differs in “hot” versus “cold” real estate markets. A paper that deals with all four biases is Fisher et al. (2003).

³⁴ See Geltner (1991) and Ross and Zisler (1991).

³⁵ See Munneke and Slade (2000).